

# Valentin Guigon

## Postdoctoral researcher in computational neuroscience, computational psychiatry, and NeuroAI

Social Learning and Decisions Lab, Department of Psychology, University of Maryland

College Park, Maryland, USA

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### RESEARCH INTERESTS

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**Themes**    Belief updating; information-seeking; social learning; metacognition; decision-making.

**Disciplines**    Computational psychiatry; computational neuroscience; behavioral economics; NeuroAI.

**Broader interests**    Public policy, epistemology, photography, and culture writ large.

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### ACADEMIC POSITIONS

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#### Postdoctoral researcher · Social Learning and Decisions Lab

March 2024 – present

##### Department of Psychology, University of Maryland

College Park, MD, USA · Social learning · PI: Caroline Charpentier

Combining computational modeling with neuroimaging to investigate the neurocomputational mechanisms underlying neurotypical and neuroatypical learning in social contexts, and individual differences in dynamic belief updating during trust learning.

Lead development of lab-wide reproducible infrastructure for behavioral, fMRI, and model-fitting workflows, including shared pipelines, QC and reporting systems.

Research assistants: Joshua Berman, Atharv A. Umap, Gaurav D. Mahajan.

#### Postdoctoral researcher · Neuroeconomics Laboratory

March 2023 – Oct 2023

##### CNRS UMR 5229, ISCMJ · CNRS UMR 5824, GATE

Lyon, France · Social decision-making · PIs: Marie Claire Villeval, Jean-Claude Dreher

Extended my doctoral work on the transmission of uncertain information to belief updating in social-network settings. The position centered on behavioral study design, implementation, analysis planning, and model-based interpretation of social influence and belief revision.

#### PhD researcher · Neuroeconomics Laboratory

Oct 2018 – Dec 2022

##### CNRS UMR 5229, ISCMJ · CNRS UMR 5824, GATE

Lyon, France · Social decision-making · PIs: Marie Claire Villeval, Jean-Claude Dreher

Designed and analyzed behavioral and neuroimaging studies on how people seek, evaluate, and transmit uncertain information in social and economic environments. The dissertation developed links between metacognition, information seeking, belief updating, and inferences about others' preferences for information.

#### Research intern · Neuroeconomics Laboratory

Feb – July 2018

##### CNRS UMR 5229, ISCMJ

Lyon, France · Decision-making and reward processing · PI: Jean-Claude Dreher

Conducted functional-connectivity analyses of testosterone-induced effects on orbitofrontal-amygdala coupling during anticipation of primary and secondary rewards, using first-level and connectivity models for fMRI data.

#### Research intern · Neuroeconomics Laboratory

June – Aug 2017

##### CNRS UMR 5229, ISCMJ

Lyon, France · Decision-making and reward processing · PI: Jean-Claude Dreher

Contributed to fMRI analyses of rewarded tasks and to research and ethics protocols for a study on moral norms under tDCS-disrupted decision-making.

#### Additional early research experience

March – April 2016

##### CNRS UMR 7260; CNRS UMR 7260; La Poste; CNRS UMR 7309

Vestibular cortex and bodily self-consciousness; complex-systems modeling (Lotka-Volterra); nudging for risky behavior and ecological transition; theory of mind in schizophrenia.

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## PUBLICATIONS

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### PEER-REVIEWED

- Benistant, J., **Guigon, V.**, Nicolas, A., Derrington, E., & Dreher, J. C. (2026). Social learning dynamically shapes moral decision-making by biasing subjective valuation. *PLoS Biology*.
- Guigon, V.**, Geay, L., & Charpentier, C. J. (2026). Rethinking misinformation through plausibility estimation and confidence calibration. *Communications Psychology*, 4, 24.
- Guigon, V.**, Villevall, M. C., & Dreher, J. C. (2024). Metacognition biases information seeking in assessing ambiguous news. *Communications Psychology*, 2(1), 122.
- Hu, Y., **Guigon, V.**, Philippe, R., Zhao, S., Derrington, E., Corgnet, B., ... & Dreher, J. C. (2022). Perturbation of Right Dorsolateral Prefrontal Cortex Makes Power Holders Less Resistant to Tempting Bribes. *Psychological Science*, 33(3), 412–423.
- O'Connor, D. A., Janet, R., **Guigon, V.**, Belle, A., Vincent, B. T., Bromberg, U., ... & Dreher, J. C. (2021). Rewards that are near increase impulsive action. *iScience*, 24(4), 102292.

### PREPRINTS

- Umap, A., A., Charpentier, C. J. & **Guigon, V.**, Neurodesign-plus: design optimization for constrained event-level fMRI paradigms. *psyRxiv*.

### UNDER REVIEW

- Guigon, V.**, Benistant, J., Villevall, M. C., & Dreher, J.-C. (2026). Neurocomputational mechanisms of inferring others' preferences for information. *Research Square*.
- Mahajan, G. D., Charpentier, C. J. & **Guigon, V.**, Ten simple rules for achieving computational reproducibility in neuroscience. *psyRxiv*.
- Mombelli, S., ..., **Guigon, V.**, ..., & Benrimoh, D. (2026). Integrating theory-driven and data-driven computational psychiatry. *psyRxiv*.

### IN PREPARATION

- Guigon, V.**, Topel, S., & Charpentier, C. J. Individual differences in dynamic belief updating during trust learning. *Conference presentations: SNE (2025); CISE (2025); PTSF (2026)*.
- Guigon, V.**, Dunne, S., Pazderska, A., Frodl, T., Nolan, J. J., Clairis, N., O'Doherty, J. P., & Dreher, J.-C. Testosterone causes decoupling of orbitofrontal cortex-amygdala relationship while anticipating primary and secondary rewards. *Conference presentation: OFC Meeting (2018)*.

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## FELLOWSHIPS, AWARDS AND AFFILIATIONS

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Founding member and Cohort 001 expert at DisCo (2026)  
UMD Department of Psychology 2026 Team Leader Open Science Award (2026)  
Fellow, Thinking About Thinking (2026–present)  
Full Member, Sigma Xi, the Scientific Research Honor Society (2026–present)  
Affiliate, Neuroscience and Cognitive Science program (NACS), UMD (2025–present)  
Affiliate, Artificial Intelligence Interdisciplinary Institute at Maryland (AIM), UMD (2024–present)  
Cortex-magazine, editorial board member (January 2023–January 2024)  
LabEX CORTEX fellowship (2022–2023)  
LabEX CORTEX fellowship (2021–2022)  
IDEX INDEPTH PhD fellowship (2018–2021)

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## SELECTED SKILLS

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| Modeling   | Computational modeling and simulation; Bayesian statistical analysis and inference; reinforcement-learning, bayesian and decision-making models; neural networks and deep learning. |
| Methods    | Experimental design; simulations; neuroimaging; tDCS; open-science and reproducibility.                                                                                             |
| Leadership | Project leadership; cross-disciplinary collaboration; teaching, mentoring, and scientific communication.                                                                            |

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## SELECTED TOOLS AND METHODS

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|------------------------|--------------------------------------------------------------------------------------------------------------|
| Authored software      | neurodesign-plus (fMRI experimental-design optimization): <a href="#">PyPI</a>   <a href="#">ReadTheDocs</a> |
| Programming            | MATLAB, Python, R                                                                                            |
| Computational modeling | Variational VBA toolbox, JAGS, BRMS, PyMC, PyTorch, Hugging Face                                             |
| Neuroimaging           | SPM, CONN, fMRIPrep, Nilearn, Fitlins   MRIcroGL, WFU Pickatlas, DANDI   IRBs                                |

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## EDUCATION

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| <b>PhD in Biology, Cognitive Neuroscience</b><br>Neurosciences and Cognition Doctoral School, Université Claude Bernard Lyon 1, France<br>Dissertation: <a href="#">Neurocognitive mechanisms underlying transmission of uncertain information in humans.</a> | 2018–2022 |
| <b>MSc in Cognitive Science</b><br>Faculty of Psychology, Université Lumière Lyon 2, France                                                                                                                                                                   | 2016–2018 |
| <b>BSc in Physiology and Neuroscience, Cognitive and Adaptive Neuroscience</b><br>Faculty of Sciences, Aix-Marseille Université, France                                                                                                                       | 2015–2016 |
| <b>BSc in Psychology</b><br>Faculty of Human Sciences, Aix-Marseille Université, France                                                                                                                                                                       | 2012–2015 |

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## TEACHING AND MENTORING

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### TEACHING

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| <b>Instructor · NACS 645 Cognitive Science</b><br>University of Maryland · 11 PhD students<br>Course: <a href="#">Foundational debates in cognitive science</a> . Topics: concepts, modularity, embodied cognition, cognitive architecture, methods, social cognition, intercultural cognition, cognitive systems, thinking, and NeuroAI. | Sept–Dec 2025                   |
| <b>Instructor · Cortecs Critical Thinking Summer School</b><br>France · 20 students<br>Course: <a href="#">Information; Beliefs; Predictions.</a>                                                                                                                                                                                         | July 2025                       |
| <b>Instructor · SPM fMRI preprocessing and contrasts</b><br>Université Claude Bernard Lyon 1 · 20 PhD students<br>Course: <a href="#">fMRI preprocessing and first-level modeling with SPM.</a>                                                                                                                                           | June 2023                       |
| <b>Teaching assistant · Cognitive Psychology</b><br>Université Lumière Lyon 2 · 6 × 40 undergrad students · Professors: François Osiurak; Gaën Plancher<br>Licence 1 and License 2 courses covering research methodology, experimental paradigms, emotions, working memory, executive functions, and language.                            | Sept–Dec 2017, 2018, 2019, 2020 |

### MENTORING

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|------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Research mentor · Research assistants</b><br>University of Maryland · 3 students                        | March 2025–ongoing   |
| <b>Research mentor · PhD student</b><br>University of Maryland · 1 student                                 | Sept 2024–April 2025 |
| <b>Research mentor · Neuroscience Master students</b><br>Université Claude Bernard Lyon 1 · 2 × 3 students | Jan–June 2023, 2024  |

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## INVITED TALKS AND SCIENCE DISSEMINATION

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### Substack: Medium

- 2026:** Invited talk: Managing Information Under Uncertainty - From Human Judgment to AI-Mediated Cognition at Google Deepmind ([Information Quality Talk Series](#)).
- 2025:** Invited article: La sphère publique et la chambre d'écho. [Cortecs.org](#).
- 2024:** Interview: S. Tremblay & CheeseNaan Productions. Au-delà des écrans : comprendre la désinformation en ligne. [Spotify](#).
- 2023:** Invited article: Notre société est-elle de plus en plus polarisée ? [Cortex-mag.net](#).
- 2021:** Invited talk: Information, Beliefs, and Knowledge at ISEG Marketing School.  
Interview: A. Sorce. L'invité de la semaine - Valentin Guigon, fake news et théories du complot. [Pharefm.com](#).  
Interview: O. Mollaret. À Lyon : les victimes de fake news "fonctionnent comme tout le monde". [Rue89lyon.fr](#).  
Interview: Yanis. Comment ne pas se faire avoir par les fake news ? [Alveole.media / Cognitif](#).
- 2020:** Interview: A. Gabert. Pourquoi les théories du complot plaisent à notre cerveau. [Cortex-mag.net](#).

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## EXTERNAL TRAINING

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- 2026:** [NeuroAI](#), Neuromatch, 85h (Certified);
- 2025:** [Hugging Face AI Agents](#), 28h.
- 2024:** [Hugging Face Deep RL](#), Hugging Face, 28h (Certified); [Reproducible Research II](#), INRIA (FUN-MOOC), 35h; [Reproducible Research](#), INRIA (FUN-MOOC), 24h (Certified); [ML with scikit-learn](#), INRIA (FUN-MOOC), 36h (Certified); [Structural Equation Modeling](#), CenterStat, 18h (Certified); [Game Theory](#), Stanford University (Coursera), 17h.
- 2023:** [Hugging Face NLP Course](#), Hugging Face, 49h; [CS50](#), Harvard University (edX), 30h; [ML with Python](#), IBM (Coursera), 15h.
- 2022:** [Bayesian Statistics in Evolutionary Biology](#), LBBE (UCBL1), 27h (Certified).
- 2021:** [Statistical Bayesian Modelling](#), L. Nalborczyk (MaiMoSiNe), 20h (Certified); [Integrity and Ethics in Research Careers](#), UDL (FUN-MOOC), 30h (Certified).
- 2020:** [Model Thinking](#), S. E. Page (Coursera), 27h.
- 2019:** [Fundamentals of Noninvasive Brain Stimulation](#), BrainBox, 14h (Certified); [Leaders of Learning](#), Harvard University (FUN-MOOC), 15h (Certified); [Experimental Economics](#), J. Rosaz (GATE Lab), 20h. [Behavioral Economics](#), B. Corgnet, A. Zylbersztejn (GATE Lab), 20h. [Introduction to Neuroeconomics](#), V. Klucharev (Coursera), 31h.
- 2018:** [Principles of fMRI I and II](#), M. Lindquist & T. Wager (Coursera), 17h.